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The impact of gamified interactive e-books incorporating metacognitive reading strategies on Chinese elementary students' mathematical reading comprehension, word problem-solving performance, and general reading motivation

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Abstract
Mathematical word problem performance is a crucial indicator of students' mathematical abilities, and enhancing mathematical reading comprehension and reading motivation is key. This study examines the impact of a gamified interactive e-book (GIEB) that incorporates metacognitive reading strategies on reading motivation, mathematical reading comprehension, and word problem performance among fifth-grade elementary students. Using a quasi-experimental design, the experimental group learned through the GIEB, while the control group received traditional metacognitive reading strategy instruction. The results show that the experimental group significantly outperformed the control group in reading motivation, mathematical reading comprehension, and word problem performance. GIEB, through strategies such as prediction, monitoring, and reflection, improved students' ability to analyze information, while its integration of immediate feedback and engaging design effectively stimulated students' intrinsic motivation to learn, strengthening their interest and focus on learning. This study confirms the critical role of reading motivation and mathematical reading comprehension in word problem learning and supports the application of Self-Determination Theory in education. It provides practical evidence for the use of digital tools to enhance educational quality and aligns with the United Nations Sustainable Development Goal 4, promoting the development of equitable and high-quality education. However, the generalizability of the findings may be limited to Chinese elementary school students, and future research is needed to validate the results in other cultural or educational contexts.

Extended author information available on the last page of the article

Keywords Gamification · Gamified interactive e-book · Metacognitive reading strategies · Reading motivation · Mathematical reading comprehension · Mathematical word problems

1 Introduction

1.1 Background and importance of mathematical word problems

With the advancement of digital and mobile technologies, mathematics has become a foundational element of modern civilization, as digital systems fundamentally rely on mathematical principles (Muranov et al., 2023). It is widely acknowledged that mathematical literacy is essential for addressing real-world problems (Akhter & Akhter, 2018; Chew et al., 2019), and early education plays a crucial role in equipping students with the skills needed for success in more advanced mathematics (Hwang et al., 2021).

The 2030 Agenda for Sustainable Development by the United Nations underscores this point in Goal 4 (SDG4; Quality Education), calling for global improvements in literacy and numeracy. Numeracy is not limited to basic arithmetic—it also encompasses the ability to interpret and reason with quantitative information (Mahmud & Pratiwi, 2019). Similarly, literacy goes beyond decoding text to include comprehension, critical thinking, and communication (Telaumbanua et al., 2024). These two domains, often considered separately, are inherently interconnected, especially in mathematics education where students frequently engage with word-based mathematical problems (Aguila, 2020).

As education systems strive to prepare students for STEM pathways, the development of problem-solving skills becomes increasingly important (Beal & Galan, 2015). The ability to solve problems is a critical indicator of mathematical competence, requiring students to gain experience in transferring and applying their knowledge and skills to real-world problems during the learning process (Permata et al., 2018). Among mathematical problem-solving tasks, word problems are particularly versatile, covering a wide range of applications—from simple everyday scenarios to complex scientific calculations (Moleko & Mosimege, 2020). However, solving word problems is highly challenging, as it simultaneously tests both mathematical proficiency and text reading and comprehension skills (Schumacher & Fuchs, 2012).

MWPs are defined as tasks that convey information through language rather than symbols (Timmermans et al., 2007), requiring students to comprehend and translate text into mathematical expressions. Table 1 shows the difference between Mathematical Word Problems (MWPs) and Mathematical Symbols. This comparison demonstrates that MWPs require more than just arithmetic skills that necessitate the ability to interpret and translate language into mathematical operations. This is one of the reasons why metacognitive reading strategies are essential for solving MWPs effectively. To enhance students' mathematical abilities and achieve the aforementioned Sustainable Development Goals, educators should prioritize improving students' performance on mathematical word problems. This effort should begin during elementary education, where foundational content is introduced. Building a strong

Table 1 The difference between mathematical word problems (MWPs) and ordinary mathematical symbols

Aspect	Mathematical Word Problems (MWPs)	Mathematical Symbols
Definition	Problems presented in textual form, requiring comprehension and translation into mathematical expressions.	Problems directly expressed using mathematical symbols and operations.
Example	"Jane has 5 apples. She buys 3 more. How many apples does she have in total?"	5 + 3=?
Focus	Emphasizes reading comprehension and the ability to extract relevant information.	Focuses purely on computational skills and applying mathematical rules.
Required Skills	Requires both linguistic and mathematical understanding to identify operations and relationships.	Requires familiarity with mathematical notation and operations.
Real-world Context	Often framed in a relatable scenario (e.g., buying items, traveling distances).	Abstract representation, without contextual details.
Example with Extra Details	"A bus starts with 10 passengers. At the first stop, 4 passengers get off, and 6 more get on. How many passengers are on the bus now?"	10 − 4 + 6=?

mathematical foundation at this stage not only strengthens problem-solving skills but also fosters interdisciplinary competencies and cultivates lifelong learning habits (Hattie, 2012; Rodriguez-Jimenez et al., 2023).

In particular, China's elementary education system is undergoing significant curriculum reforms that prioritize core competencies, problem-solving, and application-oriented learning from an early age (Communist Party of China Central Committee & State Council, 2025). These changes reflect a broader global trend toward reducing rote learning and promoting deeper cognitive engagement (Tursynkulova et al., 2023; Weng et al., 2023). Examining how Chinese students respond to complex tasks such as mathematical word problems (MWPs) within this evolving context offers insights into national policy implementation and provides implications for other systems undergoing similar educational transformations. One key focus of these reforms is MWPs, which require the integration of reading comprehension and mathematical reasoning.

1.2 Challenges in solving MWPs and the role of reading comprehension

Due to their complexity, MWPs are widely recognized as one of the most challenging forms of mathematical tasks for elementary students (Morton & Qu, 2013; Şahinkaya et al., 2024; Swanson et al., 2024; Smeltzer et al., 2024). To address this, the Mathematics Curriculum Standards for Compulsory Education (2022 Edition) in China

emphasize fostering students' application awareness through MWPs (Ministry of Education of the People's Republic of China, 2022). Despite this policy support, Chinese elementary students still encounter notable difficulties with MWPs, which can result in decreased motivation and hinder future academic achievement (Guan, 2023; Zhang, 2023; Smeltzer et al., 2024; Vilenius-Tuohimaa et al., 2008).

One key issue is students' inadequate reading comprehension abilities. Comprehension involves understanding written materials, identifying important information, and forming mental models of problems (Lin et al., 2021; Ayça, 2022). Many students fail to distinguish relevant details from extraneous information and often attempt to mechanically convert words into calculations without understanding the problem's structure (Vula et al., 2017). A strong body of research has demonstrated that reading comprehension significantly correlates with MWP performance (Vilenius-Tuohimaa et al., 2008; Basol et al., 2011; Boonen et al., 2016). However, this skill is often neglected in mathematics instruction, particularly in China (Mulyati et al., 2017; Zhang, 2023).

1.3 Metacognitive reading strategies: potentials and limitations

To address this gap, metacognitive reading strategies—methods that help learners plan, monitor, and evaluate their understanding—have been widely recommended (Phakiti, 2003; Meniado, 2016). These strategies improve reading comprehension and are increasingly being explored in mathematics education (Channa et al., 2015; Ali & Razali, 2019; Ghadamgahi & Ghafournia, 2022; Mevarech et al., 2010; Yang, 2012; Vula et al., 2017). Nonetheless, some studies have found these strategies to be less effective in certain contexts, especially when learners find the materials unengaging or overly repetitive (Vauras et al., 1999; Meniado, 2016; Arif et al., 2020; Tedjo et al., 2022).

Learner motivation is a key factor influencing the success of metacognitive strategy use. Emotional states such as boredom have been found to reduce students' willingness to apply reading strategies and impair comprehension (La Marca & Longo, 2017; Shehzad et al., 2020). On the other hand, reading motivation has been positively linked to comprehension and mathematical achievement (Pollack et al., 2021; Tian et al., 2018; Wigfield et al., 2016).

1.4 Technology, gamification, and the present study

Given the increasing availability of digital tools, educators now have opportunities to design more engaging and motivating learning environments. The integration of technology into classroom practices supports students' engagement, creativity, and exploration (Geiger et al., 2010; Muranov et al., 2023). The Ministry of Education in China also supports the integration of digital technologies into teaching practices to improve the quality of mathematics education (Ministry of Education of the People's Republic of China, 2022). Interactive e-books, in particular, offer multimedia and interactive elements that can enhance reading experiences and learning outcomes (Kong et al., 2018; Bus et al., 2019).

Gamification—the application of game design elements in non-game educational contexts—has also gained traction as a strategy for increasing learner engagement and motivation (Deterding et al., 2011; Yan et al., 2023). It can be combined with digital resources like interactive e-books to create immersive, motivating learning environments that encourage sustained participation and academic effort (Baxter et al., 2015; Zainuddin et al., 2020; Jutin & Maat, 2024; Qomario et al., 2020).

Given these opportunities, the present study aims to design and evaluate a gamified interactive e-book that incorporates metacognitive reading strategies. The goal is to enhance Chinese elementary students' motivation, reading comprehension, and ultimately their performance in mathematical word problems. The study is guided by the following research questions:

1. Does the gamified interactive e-book incorporating metacognitive reading strategies enhance the reading motivation of Chinese elementary school students more effectively than traditional metacognitive reading strategies?
2. Does the gamified interactive e-book incorporating metacognitive reading strategies improve the reading comprehension skills of Chinese elementary school students more effectively than traditional metacognitive reading strategies?
3. Does the gamified interactive e-book incorporating metacognitive reading strategies improve the performance of Chinese elementary school students in solving mathematical word problems more effectively than traditional metacognitive reading strategies?

2 Literature review

2.1 Metacognitive reading strategies

Metacognitive reading strategies refer to planned, intentional, goal-directed, and future-oriented mental activities or processes that help learners reflect on and assess how they complete cognitive tasks (Meniado, 2016). These strategies aim to enhance learners' awareness and control, improve reading comprehension skills, and determine whether they have exerted their best effort (Zhang & Seepho, 2013). Metacognitive reading strategies can be categorized into three phases: **planning** (pre-reading), **monitoring** (during reading), and **evaluation** (post-reading), each involving specific strategies that require metacognitive processing by the reader (Iwai, 2011).

Planning involves selecting appropriate strategies and allocating resources that influence performance, integrating predictions about developments with preparatory responses to potential scenarios (Zareei, 2007). **Monitoring** is a strategy that analyzes information as tasks progress (Meniado, 2016). **Evaluation** is defined as assessing the conclusions of personal learning and regulating processes (Meniado, 2016).

Metacognitive reading strategies provide educators with a broader framework, emphasizing the three-step process of planning, monitoring, and evaluation during reading. Educators must flexibly combine various strategies to adapt to different learning tasks and objectives.

Recent review studies on metacognitive reading strategies indicate their widespread application in language learning and their effectiveness in enhancing students' reading comprehension skills (Ali & Razali, 2019; Kan et al., 2024). Metacognitive reading strategies foster learners' metacognitive awareness, enabling them to recognize their cognitive processes, address reading challenges, and ultimately improve their reading comprehension (Kan et al., 2024).

Dougherty (1996) investigated the impact of metacognitive reading strategies on the reading comprehension of third-grade students. The findings revealed that these strategies significantly enhanced students' reading comprehension. Dougherty emphasized the importance of helping students build strategy models and increasing interaction during strategy use in teaching metacognitive reading strategies. These approaches transformed students into self-regulated learners, boosted their motivation, and consequently improved their reading comprehension.

Hossu (2020) examined the relationship between the use of metacognitive reading strategies and the reading comprehension of 85 fourth-grade students. The study showed a significant positive correlation between the use of these strategies and students' reading comprehension. Hossu argued that metacognitive reading strategies should not be limited to older students; instead, students should begin learning them at the early stages of education.

Some earlier studies also suggested that metacognitive reading strategies could enhance learners' reading motivation (Öztürk & Aydogmus, 2021; Bozgun & Can, 2023). For instance, Jamshidi et al. (2013) explored the impact of metacognitive reading strategies on the reading motivation of 62 Iranian female English learners aged 18 to 30. The results indicated a significant increase in learners' reading motivation. Jamshidi et al. attributed this improvement to a learner-centered approach in the teaching of metacognitive reading strategies, which maximized learners' autonomy.

However, as research into metacognitive reading strategies has deepened, some studies suggest that their effectiveness may not be as significant as previously thought. Meniado (2016) investigated whether a definitive relationship exists between metacognitive reading strategies, reading motivation, and reading comprehension. The results indicated that students performed below average in reading comprehension, and no significant correlation was found between the use of metacognitive strategies and reading comprehension. This finding contradicted most prior research.

Meniado (2016) further observed that when students encountered lengthy and unengaging texts during metacognitive strategy training, they often abandoned reading and comprehension altogether, resorting to guessing the correct answers. The study emphasized the need to create engaging reading environments in future strategy training.

Similarly, Pei (2014) examined the effects of metacognitive reading strategies on the reading comprehension and metacognitive awareness of Chinese English learners. The findings aligned with those of Meniado (2016), showing no significant improvement in students' reading comprehension. In post-study interviews, students reported that the content taught during metacognitive strategy instruction was perceived as overly simplistic and uninteresting. They described the entire process as boring and ineffective, failing to evoke any interest or enthusiasm.

Arif et al. (2020) investigated the impact of metacognitive reading strategies on the English reading comprehension performance of Malaysian Chinese students. The results showed that these strategies did not improve students' reading comprehension. The researchers attributed this to factors such as students' behavior, attitudes, or emotions, which likely reduced their motivation. They emphasized that future metacognitive strategy training should incorporate encouragement and guidance to support students. Similarly, Ruvalcabar-Estrada et al. (2021) found that metacognitive reading strategies were not effective in enhancing students' reading motivation.

Current research on the effects of metacognitive reading strategies on mathematical word problem-solving abilities remains scarce, and their effectiveness is still inconclusive. Strohmaier et al. (2022) explored the relationship between metacognitive reading strategies and the ability to solve mathematical word problems. Their findings indicated a certain degree of correlation, suggesting that these strategies can assist students in solving mathematical word problems.

However, the study also revealed that this correlation is influenced by confounding factors such as motivation and reading comprehension skills. Strohmaier et al. (2022) concluded that relying solely on metacognitive reading strategies to improve performance in mathematical word problems may have limited effectiveness. They emphasized the need for further longitudinal studies to validate their efficacy and recommended integrating metacognitive reading strategies with other methods while taking into account additional relevant learning factors.

Our literature review indicates that while many prior studies have suggested that metacognitive reading strategies can effectively enhance students' reading comprehension and motivation (Dougherty, 1996; Jamshidi et al., 2013; Ali & Razali, 2019; Hossu, 2020; Kan et al., 2024; Öztürk & Aydogmus, 2021; Bozgun & Can, 2023), other research has questioned their effectiveness (Meniado, 2016; Arif et al., 2020; Ruvalcabar-Estrada et al., 2021; Strohmaier et al., 2022). These studies have highlighted several issues:

1. Metacognitive reading strategies have been predominantly applied in the field of language learning, with limited research focusing on their application in mathematics (Ali & Razali, 2019; Kan et al., 2024).
2. Most existing research is concentrated on higher education levels, while studies in primary education are scarce, and there is a notable lack of research in China (Kan et al., 2024).
3. The effectiveness of metacognitive reading strategies remains uncertain, necessitating further longitudinal empirical studies to validate their efficacy (Strohmaier et al., 2022).
4. There is a lack of studies integrating digital technologies with metacognitive reading strategies. Strohmaier et al. (2022) also recommended combining these strategies with other approaches.

At the same time, prior studies have provided valuable insights that inform the conceptual foundation of the present research. Researchers have highlighted the importance of enhancing interaction and helping students build strategic models during metacognitive reading strategy instruction. Supporting students' autonomy and moti-

vation has also been found essential for maximizing the impact of these strategies (Dougherty, 1996; Jamshidi et al., 2013). In contrast, studies have shown that negative emotional states such as boredom may undermine students' willingness to apply strategies effectively, especially when the learning materials are perceived as uninteresting or overly repetitive (Meniado, 2016; Arif et al., 2020). These findings point to the need for more engaging and motivating instructional designs when applying metacognitive reading strategies, particularly in elementary education settings.

2.2 Gamification

Integrating gamification with educational content and knowledge has emerged as an engaging and motivational concept for learners in an era dominated by interactive media and games (McGonigal, 2012). Gamification refers to the process of incorporating game elements and mechanisms into non-game contexts. Its primary goal is to enhance user engagement, thereby improving participation and performance (Deterding et al., 2011). The concept of gamification involves removing the foundational aspects of games while retaining their core structural elements that drive interest and engagement (Seaborn & Fels, 2015).

Gamification elements are the central components of the gamification concept, encompassing all elements and factors essential for designing and understanding gamified experiences (Schöbel et al., 2020). Hamari et al. (2014) summarized commonly used gamification elements, including points, badges, leaderboards, levels, narratives, clear goals, feedback, rewards, progress indicators, and challenges. In recent years, researchers have explored integrating new gamification elements into gamified learning. These elements aim to enhance existing learning strategies, foster user engagement, address challenges, and achieve desired learning outcomes (Zainuddin et al., 2020).

Prados Sánchez et al. (2023) investigated the impact of gamified platforms on reading comprehension and attitudes in primary education. Their findings indicated that gamified platforms significantly improved students' reading comprehension, making the reading experience more enjoyable and engaging, while also increasing students' interest and motivation. Gamification elements such as points, rewards, badges, leaderboards, progress bars, and contextualized tasks were identified as particularly effective in this study.

Ndegwa et al. (2024) examined the influence of gamified applications on reading comprehension and learning attitudes in the context of learning local languages among Kenyan primary school students. The results demonstrated significant improvements in both reading comprehension and learning motivation. Similarly, Tsai et al. (2020) developed a web-based collaborative reading annotation system (WCRAS) incorporating gamified mechanisms to enhance the reading comprehension skills of elementary school students in Taiwan. The study revealed that WCRAS effectively improved students' reading comprehension. Tsai et al. (2020) suggested that future research should provide more detailed feedback to help learners better understand texts.

Li and Chu (2021) designed a gamified reading application to improve reading comprehension and motivation among elementary school students in Hong Kong.

Their study found significant improvements in both areas. However, they noted that some students were not fond of highly competitive gamification elements, such as leaderboards or points. To address this, they recommended incorporating other gamification elements, such as collaboration and narratives, to engage students who dislike competitive features. Li and Chu (2021) also emphasized the importance of fostering intrinsic motivation in students.

Sun-Lin and Chiou (2019) developed a gamified learning system incorporating comparative strategies to enhance algebra word problem-solving performance and learning attitudes among Taiwanese elementary school students. Their findings showed significant improvements in both performance and attitudes. They argued that using gamification components such as context, progress indicators, real-time feedback, guiding prompts, challenges, and rewards could effectively boost students' learning outcomes and attitudes. Furthermore, Sun-Lin and Chiou (2019) proposed exploring the integration of metacognitive strategies with gamification in future studies.

Although numerous empirical studies have demonstrated the effectiveness of gamification strategies, they do have certain limitations. Chen et al. (2020) found that while gamified reading systems improved elementary students' reading experiences and social interactions, no significant differences in reading comprehension were observed between the experimental and control groups. They suggested that this might be due to a lack of connection between gamified design and the process of reading comprehension. Chen et al. (2020) emphasized the importance of integrating learning objectives and corresponding gamification elements into the design to achieve positive learning outcomes.

Similarly, García-Iruela et al. (2022) observed that gamified virtual classrooms did not significantly enhance students' motivation or engagement. They attributed this to the simplistic gamification design and the limited use of gamification elements. Kyewski and Krämer (2018) also highlighted that relying solely on basic gamification components, such as points and leaderboards, may not effectively boost students' motivation or learning outcomes.

Some scholars have raised additional concerns. For instance, Faure-Carvallo et al. (2022) noted that excessive use of gamified applications could negatively impact eyesight. Furthermore, others have argued that while gamification strategies hold promise in educational contexts, traditional teaching methods may slow down the pace at which students process and analyze information, allowing for deeper understanding and assimilation of knowledge, ultimately leading to more effective learning outcomes (Zhukova et al., 2023; Hung, 2018).

The literature review indicates that gamification effectively enhances students' motivation and academic performance (Sun-Lin & Chiou, 2019; Tsai et al., 2020; Li & Chu, 2021; Prados Sánchez et al., 2023; Ndegwa et al., 2024). However, empirical studies exploring the improvement of students' mathematical word problem-solving performance through either gamification or metacognitive reading strategies remain scarce. Moreover, no research has been identified that combines these two approaches.

Although students may experience negative emotions, such as boredom, during the training process of metacognitive reading strategies, potentially affecting their

motivation and learning outcomes, these strategies provide a systematic learning method and reflective framework to improve reading comprehension (Meniado, 2016; Arif et al., 2020). Gamification, on the other hand, can offer an engaging learning environment that boosts motivation, potentially addressing the shortcomings of metacognitive reading strategies. This combination may more effectively enhance students' mathematical word problem-solving performance, offering an alternative pathway to achieving sustainable development goals.

Previous studies also provide valuable insights for effective gamification design:

1. Strengthen the connection between gamification and reading comprehension by integrating learning objectives and corresponding gamification elements into the design (Chen et al., 2020).
2. Focus on cultivating students' intrinsic motivation through gamification (Li & Chu, 2021).
3. Avoid relying solely on basic gamification components, as these may not effectively enhance students' motivation and learning performance. Additionally, include elements suitable for students who may dislike highly competitive gamification features (Kyewski & Krämer, 2018; García-Iruela et al., 2022; Li & Chu, 2021).

In this study, we selected the fifth-grade mathematics curriculum in China as the content focus and developed a Gamified Interactive E-Book (hereafter referred to as GIEB) that integrates metacognitive reading strategies. This GIEB aims to improve Chinese elementary students' reading motivation, reading comprehension skills, and mathematical word problem-solving performance.

3 The GIEB design process

3.1 System architecture of GIEB

The gamified interactive e-book (GIEB) was developed using the "Yoya" platform by Xiamen Yoya Network Technology Co., Ltd., which supports multimedia integration, interactive assessment design, and animated instruction. The GIEB aims to enhance Chinese elementary students' reading motivation, reading comprehension, and mathematical word problem-solving abilities by incorporating metacognitive reading strategies within a gamified framework.

The GIEB system consists of three core modules: the **Learning Content Module**, the **Interactive Module**, and the **Gamified Learning Module** (see Fig. 1).

The **Learning Content Module** presents mathematics material aligned with China's fifth-grade curriculum. It embeds metacognitive prompts into reading tasks to guide students through the processes of planning, monitoring, and evaluation, thus scaffolding strategic reading and comprehension.

The **Interactive Module** promotes student engagement through real-time feedback, embedded quizzes, and adaptive assistance. It supports learners in applying metacognitive strategies and responding to reading challenges dynamically.

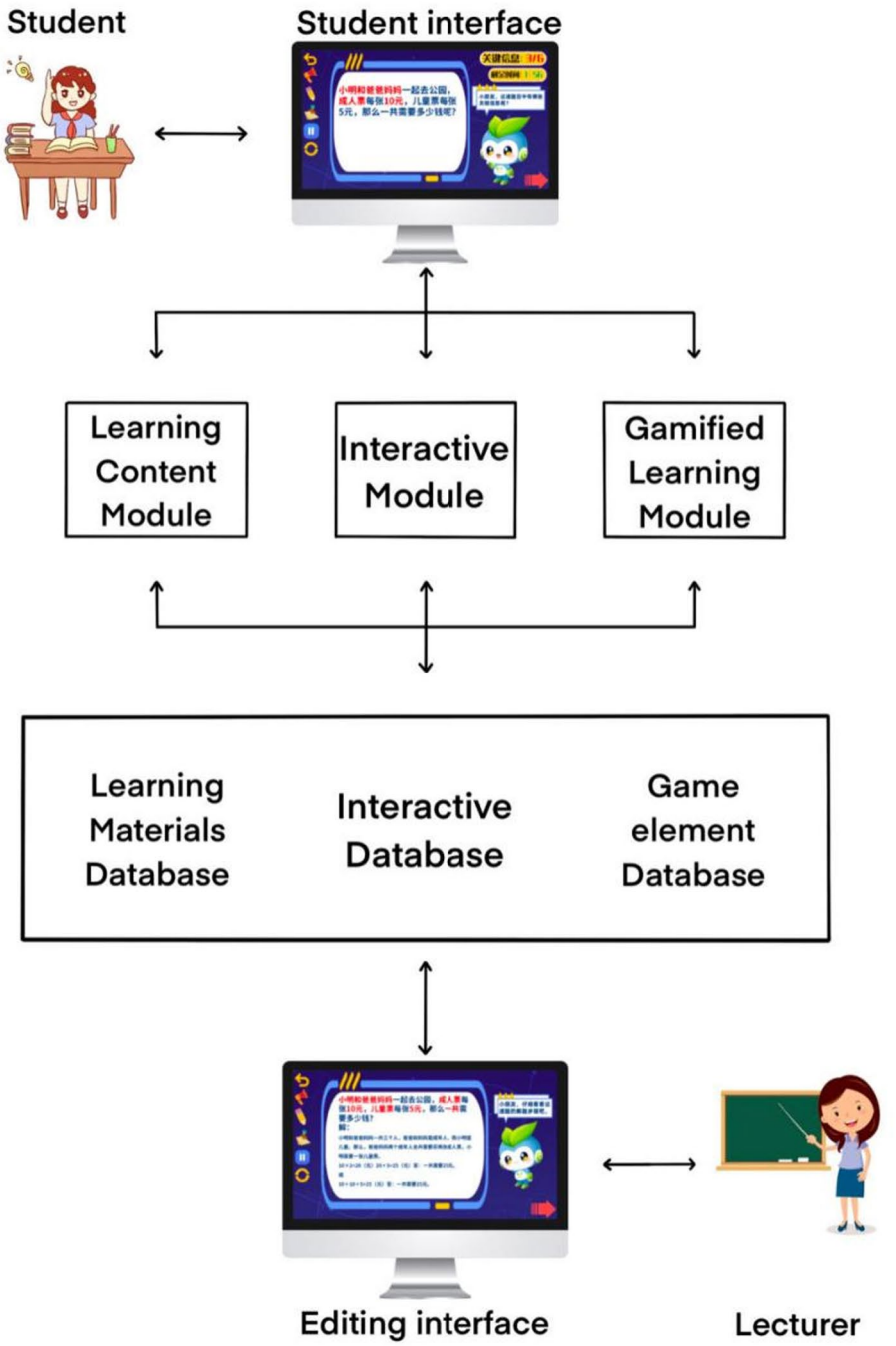


Fig. 1 System architecture of GIEB

The **Gamified Learning Module** incorporates game elements such as scenario-based tasks, points, and motivational feedback mechanisms. These features are designed to satisfy students' psychological needs for autonomy, competence, and relatedness—key components of Self-Determination Theory—thereby fostering intrinsic motivation and sustained engagement.

This modular design combines metacognitive strategy instruction with interactive and gamified features, offering a coherent, motivating, and learner-centered environment to support performance in mathematical word problems.

3.2 Theoretical framework of the GIEB

This study designs a gamified interactive e-book (GIEB) incorporating metacognitive reading strategies grounded in Self-Determination Theory (SDT), with the goal of enhancing students' motivation and learning performance by fulfilling their basic psychological needs for autonomy, competence, and relatedness (Vallerand, 2000). SDT was chosen as the guiding theoretical framework due to its robust empirical foundation in educational psychology and its clear applicability to designing learning environments that support psychological needs—particularly within gamified and student-centered contexts (Deci & Ryan, 2000; Reeve, 2012). According to SDT, intrinsic motivation flourishes when these three needs are met, promoting deeper engagement in learning and cognitive development (Donohoe & O'Sullivan, 2015; Rončević Zubković et al., 2022). In the context of educational gamification Sailer and Homner (2020a, b) demonstrated that aligning design features with these needs can significantly enhance learner motivation. Similarly, Li and Chu (2021) emphasized the critical role of intrinsic motivation in improving reading comprehension and engagement.

Building upon SDT, Kam and Umar (2018) proposed a gamified design framework that systematically maps gamification strategies to the satisfaction of autonomy, competence, and relatedness (see Fig. 2). This framework informs the structure and features of the GIEB developed in this study.

To support autonomy, the framework recommends providing meaningful choices, opportunities for repeated attempts, and learning paths tailored to students' preferences, while minimizing the penalization of task failure. To enhance competence, the framework emphasizes matching tasks to learners' ability levels and delivering

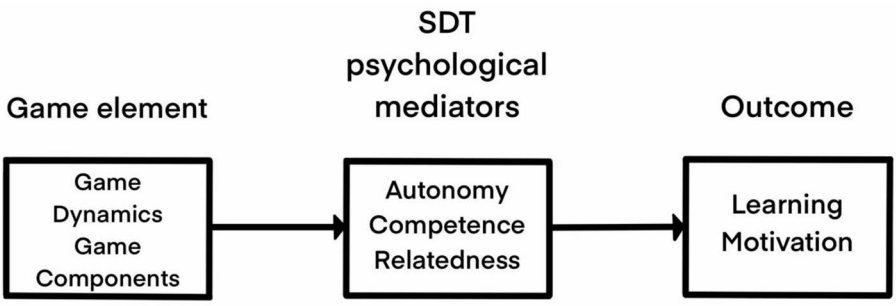


Fig. 2 SDT-based design framework for gamification

timely, constructive feedback that helps students recognize their progress. Finally, to promote relatedness, it encourages interaction with peers and learning content—for instance, through in-app discussions or personalized achievement displays—to foster a sense of connection and belonging.

By aligning the design of the GIEB with these principles, the present study aims to not only support students' psychological needs but also create a more motivating and cognitively engaging learning experience.

3.3 Development phases of the GIEB

When students use the Gamified Interactive E-book (GIEB), each learning session follows a structured sequence of three stages: Pre-Reading (Planning), Mid-Reading (Monitoring), and Post-Reading (Evaluating). These phases align with the metacognitive reading strategies of prediction, self-questioning, and reflection, respectively (see Fig. 3).

Across all three stages, the GIEB integrates three functional modules—learning content, interactive, and gamified learning—to provide comprehensive cognitive and motivational support. The learning content module delivers contextualized word problems and instructional materials; the interactive module facilitates comprehension through scaffolded engagement; and the gamified module promotes sustained motivation via game elements tailored to each phase.

This coordinated structure ensures that each metacognitive strategy is effectively operationalized through targeted content delivery, learner interaction, and motivational design.

Pre-reading (Planning) stage: prediction strategy In the pre-reading stage, the prediction strategy is applied to help students anticipate the main issues within mathematical application problems. Students begin by selecting a learning topic, after which the system presents a contextualized word problem via the learning content module.

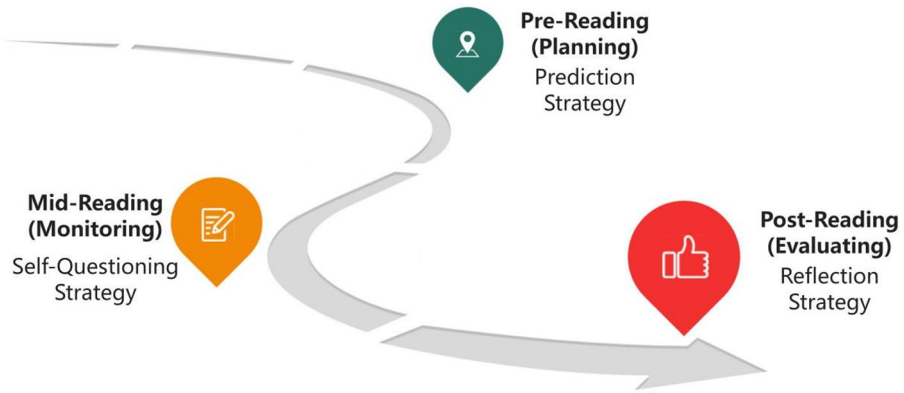


Fig. 3 The application of metacognitive reading strategies at different stages

At the start of the lesson, the virtual assistant prompts students to read the problem text carefully and identify key elements such as numbers, units, and keywords. To reinforce this process, the gamified module initiates an "Information Search" task that requires students to highlight key information within a set time limit using the GIEB's annotation tool (see Fig. 4). The virtual assistant indicates the total number of items to be found, and students who complete the task on time receive points and badges; those who exceed the time limit receive no reward (see Fig. 5).

After submission, students are guided by the assistant to predict the likely mathematical operation required (e.g., addition, subtraction, multiplication, or division) based on their identified information (see Fig. 6). This structured sequence encourages students to focus on essential details, activate relevant prior knowledge, and construct an initial understanding of the problem's nature.

Mid-reading (Monitoring) stage: self-questioning strategy In the mid-reading stage, the self-questioning strategy is implemented to help students verify their understanding. The GIEB first presents the correct key information from the text for comparison (see Fig. 7). Students who accurately identified the information receive points or badges, while those who did not receive no reward (see Fig. 8).

Next, the virtual assistant prompts students to consider the main task and the type of mathematical operation involved, then submit their responses. The system reveals the correct answer and provides immediate feedback through the same reward mechanism.



Fig. 4 The "Information Search" task



Fig. 5 Rewards for task completion

Finally, students complete and submit their solutions within a designated time frame, with rewards granted for timely responses only. This structured process reinforces comprehension and progressively strengthens students' self-monitoring capabilities.

Post-reading (Evaluating) stage: reflection strategy In the post-reading stage, the reflection strategy is applied to prompt students to evaluate the accuracy of their responses. The system informs students whether their submitted answer was correct; correct responses earn points and badges, while incorrect ones receive no reward. For incorrect answers, the GIEB presents the correct solution along with step-by-step explanations to support reflective learning (see Fig. 9). This process encourages students to internalize problem-solving methods, build confidence, and enhance their mathematical reasoning skills.

After completing the core lesson, students can independently select additional problems from the application problem bank for extended practice. Although the system no longer provides feedback on highlighted information in this mode, students may continue to mark what they consider key elements and receive points for correct answers. To foster motivation and peer interaction, GIEB also allows students to share their scores and badges within the learning community.

Table 2 summarizes the key functions of the GIEB and how they were implemented using the Yoya platform:



Fig. 6 Prediction of mathematical operation types

3.4 Game elements in GIEB

This study leverages gamification to enhance students' motivation and engagement in learning metacognitive reading strategies. Research shows that incorporating game elements into educational content can significantly improve motivation and learning outcomes (Mehta et al., 2022).

Grounded in Self-Determination Theory (SDT) (Deci & Ryan, 2012), which identifies autonomy, competence, and relatedness as three fundamental psychological needs, the GIEB design adopts the gamification framework proposed by Kam and Umar (2018) to fulfill these needs and promote intrinsic motivation.

Autonomy is supported by allowing students to choose mathematical topics freely and navigate content at their own pace. Learners can start, pause, repeat, or skip sections based on individual preferences. Tasks can be attempted multiple times without penalty, enabling self-directed learning. These features align with Buckley & Doyle's (2017) emphasis on flexible gamification designs that accommodate learner control and choice.

Competence is fostered through scaffolded problem-solving, metacognitive prompts, and real-time feedback. Each correct response earns points, which can be exchanged for badges, while incorrect responses trigger explanatory feedback. A virtual assistant provides ongoing guidance, helping students gradually master strategy



Fig. 7 Display of correct key information

use. Time constraints are also integrated to simulate authentic problem-solving scenarios and enhance students' sense of accomplishment.

Relatedness is encouraged by enabling students to share their badges and scores with peers, fostering community, collaboration, and mutual support.

By integrating gamified mechanics with metacognitive strategy instruction and contextualized problem texts, the GIEB creates an engaging and supportive learning environment. This SDT-aligned design enhances students' intrinsic motivation, promotes active use of metacognitive reading strategies, and ultimately aims to improve their mathematical reading comprehension and word problem-solving performance.

Table 3 summarizes the gamification elements applied in the GIEB, based on autonomy, competence, and relatedness.

4 Methods

4.1 Research design

This study employs a quasi-experimental pretest/posttest design to examine the impact of a gamified interactive e-book (GIEB) incorporating metacognitive reading strategies on Chinese elementary school students' reading motivation, mathematical reading comprehension, and performance in solving word problems. Participants were randomly assigned to either the experimental group or the control group, but they were not informed of their group assignments to minimize the potential influence of the "Hawthorne Effect" (Campbell & Stanley, 2015). The experimental



Fig. 8 Rewards for successfully identifying key information

group received the GIEB intervention, while the control group underwent traditional instruction in metacognitive reading strategies. Both groups completed pretests and posttests to measure the target variables. During data analysis, pretest scores were included as covariates to account for any initial differences between groups.

4.2 Participants

The sample for this study consisted of 60 fifth-grade students (typically aged 10 to 11) from two different classes at a public elementary school in Hefei, Anhui Province, China. The school was selected using convenience sampling. Prior to the start of the study, informed consent was obtained from the parents or legal guardians of all participating students. Students were also informed of their right to withdraw from the study at any time without penalty. Subsequently, the 60 students were randomly assigned to two groups: the experimental group ($n = 30$) and the control group ($n = 30$).

4.3 Research instruments

This study employed three primary measurement tools: the Mathematics Reading Motivation Scale for Elementary Students, the Mathematics Reading Comprehension



Fig. 9 Correct answers and detailed solution steps

Test for Elementary Students, and the Mathematical Word Problem-Solving Ability Test for Elementary Students.

4.3.1 The mathematics reading motivation scale for elementary students

This scale was adapted from Wigfield & Guthrie's (1997) Motivation for Reading Questionnaire (MRQ). The Mathematics Reading Motivation Scale for Elementary Students used in this study consists of 24 items, divided into three dimensions: self-efficacy, intrinsic motivation, and extrinsic motivation. The scale employs a five-point Likert format, with response options ranging from 1 (strongly disagree) to 5 (strongly agree). Example items include: Item 4, "I am confident in understanding the specific task requirements in the reading material of mathematical word problems."; Item 12, "I enjoy reading mathematics books that contain new knowledge."; and Item 18, "I read more mathematics books to achieve better grades than my classmates."The Cronbach's alpha value of the mathematical reading motivation scale for primary school students was measured at 0.922, indicating that the scale possesses high reliability.

4.3.2 The mathematics reading comprehension test for elementary students

This test was adapted from the Upper Elementary Mathematics Reading Comprehension Test developed by Guo (2022) (Cronbach's Alpha $=0.743 > 0.7$), with permission. The original items were designed under the guidance of experienced mathematics teachers and evaluated by a team of mathematics education experts

Table 2 Key functions of the GIEB

Function	Description	Yoya's Role
Personalized Learning	Students can freely choose their learning topics, after which the system presents contextualized application problem texts. Additionally, when reviewing previously learned content, students can start, pause, repeat, or skip certain parts of the learning material at any time. They are also allowed to highlight key information.	It provides personalized content choices and creates a learning environment that is autonomous and controllable for students.
Prompting	The virtual assistant can guide students on what to do at each step, offering direction and prompts for reflection at key moments.	Based on predefined rules and content, it provides students with necessary prompts and guidance for reflection according to a set logic.
Feedback & Rewards	The system provides immediate feedback on students choices and rewards correct answers with points and badges.	Facilitates the integration of feedback mechanisms and gamified elements to enhance learning and motivation.
Social Features	Students can share their points and badges with peers.	Offers features to incorporate social elements, such as leaderboards and sharing functionalities, to promote interaction and a sense of community.

for content accuracy, linguistic clarity, difficulty level, and curriculum coverage. Revisions were made based on their feedback to ensure alignment with the study objectives.

The test includes eight types of questions: mathematical concept understanding, information filtering, reading transfer, logical reasoning, information integration, abstract generalization (10 points each), spatial imagination (15 points), and mathematical language translation (25 points). The test comprises eight items in total, with a maximum score of 100.

To assess the content validity of the adapted test, three experienced mathematics teachers with over five years of teaching experience, and familiarity with fifth-grade mathematics curricula and assessment standards, were invited to evaluate the items. Using the Scale-Content Validity Index/Average (S-CVI/Ave), the teachers independently rated each item's relevance to the measurement objectives on a four-point Likert scale (1 = not relevant, 2 = somewhat relevant, 3 = relevant, 4 = highly relevant). The S-CVI/Ave for the test was calculated as 1.0 (> 0.9), indicating excellent content validity (Polit & Beck, 2006).

Table 3 Gamification elements applied in the GIEB

Deci and Ryan SDT Theory	Gamification elements applied in the GIEB
Autonomy	**Freedom to Choose Learning Content**: Students can freely select the mathematical topics they wish to study, and the GIEB will provide corresponding contextualized word problem reading materials tailored to their choices. **Control over Navigation**: Students can start, pause, repeat, or skip sections of the interactive e-book at will.
Competence	**Challenges and Questions**: Contextualized mathematical word problems, paired with tasks aligned with metacognitive reading strategies, enable students to understand, practice, and ultimately apply these strategies in the problem-solving process. **Timely Feedback**: The system provides immediate feedback on student choices, confirming correct answers and explaining incorrect ones. **Points and Badges**: Students earn points for correct answers, which they can then redeem for badges, signifying progress and achievement. **Virtual Assistant**: The virtual assistant provides students with prompts and guidance. **Time Constraints**: Students are required to complete learning tasks within a set time limit, which helps them adapt to scenarios such as time-pressured examinations.
Relatedness	**Social Sharing**: The GIEB allows students to share their earned points and badges with their peers, encouraging interaction and friendly competition.

4.3.3 The mathematical word Problem-Solving ability test for elementary students

This test, designed to assess elementary school students' problem-solving abilities in mathematical word problems, was developed based on the fifth-grade mathematics textbooks and accompanying workbooks published by the People's Education Press, a widely used resource in Chinese primary schools. The test covers four types of word problems: numerical operations, equation solving, geometry and shapes, and comprehensive practice. Each type corresponds directly to the instructional objectives outlined in the curriculum standards.

The test items were initially drafted by the researchers and subsequently reviewed by an experienced elementary school mathematics teacher to ensure alignment with the curriculum and students' competency levels. The review process involved detailed evaluations of content accuracy, item difficulty, and linguistic clarity, with revisions made to ensure that the test items met practical teaching requirements.

The finalized test consists of eight word problems, each worth 10 points, for a total score of 80. To evaluate the test's content validity, three additional mathematics teachers with over five years of teaching experience and familiarity with the fifth-grade curriculum and assessment standards were invited to independently assess each

item. Using a four-point Likert scale (1 = not relevant, 4 = highly relevant), they rated the relevance of each item to the test's objectives. The Scale-Content Validity Index/Average (S-CVI/Ave) for the test was calculated as 1.0 (> 0.9), indicating excellent content validity (Polit & Beck, 2006). This high score reflects the strong alignment between the test items and the instructional goals.

4.4 Experimental procedure

The experiment lasted for eight weeks, encompassing different stages for the GIEB (Gamified Interactive E-Book) group and the TL (Traditional Lecture) group, as shown in Fig. 10. Before the experiment began, all students completed a pretest to assess their baseline levels in reading motivation, mathematical reading comprehension, and mathematical word problem performance, taking a total of 110 min (administered in three sessions). The pretest included a reading motivation scale, a mathematical reading comprehension test, and a mathematical word problem performance test. Following the pretest, teachers conducted a 20-minute introductory session to explain the learning objectives and experimental requirements, ensuring that all participants understood the purpose and process of the activities.

From Week 2 to Week 7, students participated in learning activities based on their assigned groups, with one teacher assigned to each group. The GIEB group engaged in 45-minute sessions using the gamified interactive e-book, in line with the standard class duration in Chinese elementary schools. The TL group, on the other hand, received traditional lectures on metacognitive reading strategies, also lasting 45 min per session. Both groups were taught the same metacognitive reading strategies (prediction, self-questioning, and reflection) and practiced with identical reading materials.

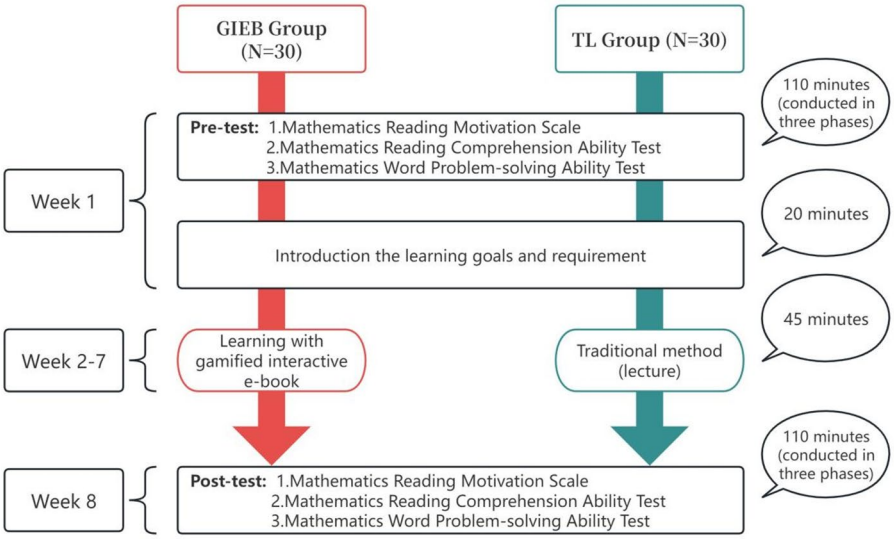


Fig. 10 Experimental procedure

In the final week (Week 8), all students completed a posttest, which also took 110 min, to evaluate changes in their reading motivation, mathematical reading comprehension, and mathematical word problem performance. The posttest included the same assessments as the pretest, including the reading motivation scale, mathematical reading comprehension, and mathematical word problem performance test.

5 Data analysis and results

To compare the effectiveness of the GIEB intervention and the traditional lecture method, this study employed Analysis of Covariance (ANCOVA). ANCOVA allows for the control of covariates, reducing error and increasing the accuracy of group comparisons. Additionally, by adjusting for initial group differences, ANCOVA enhances the reliability of the results. In this study, ANCOVA was used to analyze the differences in pretest and posttest scores between the experimental and control groups, with pretest scores included as a covariate to statistically control for existing differences. This method improves the precision and sensitivity of the analysis, making the evaluation of the intervention's effects more accurate.

Eta squared (η^2) was used to measure the impact of GIEB and the traditional lecture method on students' reading motivation, mathematical reading comprehension, and mathematical word problem performance. The use of effect size helps to mitigate the influence of sample size on the results, allowing for a better understanding of the practical significance of the experimental outcomes, rather than just determining statistical significance.

Before conducting ANCOVA, several assumptions were tested to ensure the validity of the results, including: (1) normality of data distribution, (2) homogeneity of variance, and (3) homogeneity of regression slopes. All data analyses were conducted using SPSS 26 software, with a significance level set at 0.05. Additionally, to further increase the reliability of the results, an independent samples t-test was used to compare the pretest scores of the two groups, confirming that there were no significant initial differences between the groups.

5.1 The impact of GIEB on students' reading motivation

5.1.1 Data analysis for reading motivation

To examine the impact of GIEB on students' reading motivation, several statistical analyses were conducted. First, the Shapiro-Wilk test was used to assess the normality of the pretest and posttest reading motivation scores for both the control and experimental groups. The results ($p = 0.099$, $p = 0.864$, $p = 0.452$, $p = 0.972$) indicated that the scores in both groups met the assumption of normality, as all p-values were greater than 0.05.

Next, Levene's test was performed to evaluate the homogeneity of variances across the pretest and posttest scores of the control and experimental groups. The results ($p = 0.07 > 0.05$) confirmed that the data satisfied the assumption of homogeneity of variances. Furthermore, the homogeneity of regression slopes assumption was also tested

Table 4 Independent samples t-Test results for pretest scores between experimental and control groups

Variable	Group	*N*	Mean	S.D	t	df	*p*
Pre-test	GIEB	30	72.5	14.031	0.44	58	0.661
	TL	30	74.1	14.119			

Table 5 Descriptive data and ANCOVA results for the reading motivation posttest

Variable	Group	*N*	Mean	S.D	Adjusted Mean	F	*P*	η^2
Post-test	GIEB	30	89.433	2.029	89.292	21.728	< 0.001	0.276
	TL	30	75.767	2.029	75.908			

and confirmed ($F = 0.282$, $p = 0.598 > 0.05$). These results indicated that the data met the necessary assumptions for conducting ANCOVA.

An independent samples t-test was subsequently carried out to compare the pretest reading motivation scores between the control and experimental groups, aiming to assess whether there were significant differences at baseline. The results ($p = 0.661 > 0.05$) revealed no significant differences in pretest reading motivation scores between the two groups (see Table 4).

5.1.2 Findings on reading motivation

Finally, ANCOVA was performed to examine the effect of GIEB intervention on posttest reading motivation scores while controlling for pretest scores. As shown in Table 5, the ANCOVA results revealed a significant difference in adjusted posttest scores between the two groups. The GIEB group achieved a significantly higher mean score (89.292) compared to the traditional lecture group (TL group), which scored 75.908 ($F = 21.728$, $p < 0.001$, $\eta^2 = 0.276$). These findings suggest that the GIEB intervention was more effective than traditional lecture-based instruction in enhancing students' reading motivation.

5.2 The impact of GIEB on students' mathematical reading comprehension

5.2.1 Data analysis for reading comprehension

To evaluate the impact of GIEB on students' mathematical reading comprehension, several statistical analyses were conducted. First, the Shapiro-Wilk test was used to assess the normality of the pretest and posttest scores for both the control and experimental groups. The results ($p = 0.189$, $p = 0.258$, $p = 0.786$, $p = 0.732$) indicated that the scores in both groups satisfied the assumption of normality, with all p-values exceeding 0.05.

Next, Levene's test was applied to examine the homogeneity of variances in the pretest and posttest scores between the control and experimental groups. The results ($p = 0.084 > 0.05$) confirmed that the assumption of homogeneity of variances was met. Additionally, the homogeneity of regression slopes assumption was tested and verified ($F = 0.343$, $p = 0.560 > 0.05$). These findings indicated that the data were suitable for ANCOVA.

Table 6 Independent samples t-Test results for pretest scores between experimental and control groups

Variable	Group	*N*	Mean	S.D	t	df	*p*
Pre-test	GIEB	30	52.833	6.126	0.228	58	0.821
	TL	30	53.233	7.412			

Table 7 Descriptive data and ANCOVA results for the mathematical reading comprehension posttest

Variable	Group	*N*	Mean	S.D	Adjusted Mean	F	*P*	η^2
Post-test	GIEB	30	62.633	1.067	62.761	58.930	< 0.001	0.508
	TL	30	54.700	1.067	54.572			

An independent samples t-test was conducted to compare the pretest scores of mathematical reading comprehension between the control and experimental groups, aiming to ensure baseline equivalence. The results ($p = 0.821 > 0.05$) demonstrated no significant differences in pretest scores between the two groups (see Table 6).

5.2.2 Findings on reading comprehension

Finally, ANCOVA was performed to examine the effect of the GIEB intervention on posttest scores of mathematical reading comprehension, controlling for pretest scores. As presented in Table 7, the ANCOVA results revealed a significant difference in the adjusted posttest scores between the two groups. The GIEB group achieved a significantly higher mean score (62.761) compared to the traditional lecture group (TL group), which scored 54.572 ($F = 58.930$, $p < 0.001$, $\eta^2 = 0.508$). These results suggest that the GIEB intervention was more effective than traditional lecture-based instruction in improving students' mathematical reading comprehension.

5.3 The impact of GIEB on students' performance in mathematical word problems

5.3.1 Data analysis for word problem performance

To evaluate the impact of GIEB on students' performance in mathematical word problems, several statistical analyses were conducted. First, the Shapiro-Wilk test was used to assess the normality of pretest and posttest scores for both the control and experimental groups. The results ($p = 0.306$, $p = 0.359$, $p = 0.368$, $p = 0.279$) indicated that the scores in both groups satisfied the assumption of normality, with all p-values greater than 0.05.

Next, Levene's test was performed to examine the homogeneity of variances for pretest and posttest scores between the control and experimental groups. The results ($p = 0.972 > 0.05$) confirmed that the assumption of homogeneity of variances was met. Additionally, the homogeneity of regression slopes assumption was tested and verified ($F = 2.480$, $p = 0.121 > 0.05$). These results indicated that the data were suitable for conducting ANCOVA.

An independent samples t-test was then conducted to compare the pretest scores of mathematical word problems between the control and experimental groups to ensure

Table 8 Independent samples t-Test results for pretest scores between experimental and control groups

Variable	Group	*N*	Mean	S.D	t	df	*p*
Pre-test	GIEB	30	39.6	7.820	0.157	58	0.876
	TL	30	39.933	8.618			

Table 9 Descriptive data and ANCOVA results for the performance in mathematical word problems posttest

Variable	Group	*N*	Mean	S.D	Adjusted Mean	F	*P*	η^2
Post-test	GIEB	30	47.97	0.932	47.885	29.045	< 0.001	0.338
	TL	30	40.70	0.932	40.781			

baseline equivalence. The results ($p = 0.876 > 0.05$) showed no significant differences in pretest scores between the two groups (see Table 8).

5.3.2 Findings on word problem performance

Finally, ANCOVA was conducted to investigate the effect of the GIEB intervention on posttest scores of mathematical word problems while controlling for pretest scores. As shown in Table 9, the ANCOVA results revealed a significant difference in the adjusted posttest scores between the two groups. The GIEB group achieved a significantly higher mean score (47.885) compared to the traditional lecture group (TL group), which scored 40.781 ($F = 29.045$, $p < 0.001$, $\eta^2 = 0.338$). These findings suggest that the GIEB intervention was more effective than traditional lecture-based instruction in improving students' performance in mathematical word problems.

6 Discussion

This study aimed to evaluate the effectiveness of a Gamified Interactive E-Book (GIEB) incorporating metacognitive reading strategies in enhancing Chinese elementary school students' reading motivation, mathematical reading comprehension, and performance in solving word problems. The results revealed that, compared to traditional lecture-based teaching (TL), the GIEB significantly improved students' reading motivation, mathematical reading comprehension, and word problem-solving performance. Although students taught through the TL approach demonstrated some improvement in these areas compared to their pre-test scores, the degree of improvement was not statistically significant.

6.1 Reading motivation

The GIEB intervention significantly enhanced students' reading motivation, consistent with prior studies highlighting the motivational benefits of gamified instruction (Prados Sánchez et al., 2023; Li & Chu, 2021). These benefits are largely attributed to diverse game elements that satisfy psychological needs—autonomy, competence, and relatedness—as outlined by Self-Determination Theory (SDT). In contrast, Gar-

cía-Iruela et al. (2022) argued that minimalistic gamification (e.g., simple badges or leaderboards) may be insufficient to engage learners, underscoring the importance of well-designed, multifaceted game features.

Students in the TL group showed only slight, non-significant gains in reading motivation, aligning with Ruvalcabar-Estrada et al. (2021) but diverging from Jamshidi et al. (2013). This may reflect the limitations of teacher-centered instruction, which often constrains autonomy.

The GIEB was deliberately designed to fulfill SDT's three basic needs. Autonomy was supported through choice of content and control over navigation; competence through immediate feedback, points, and virtual guidance; and relatedness through social sharing features. These elements align with findings by Sailer and Homner (2020a, b), who emphasized that meeting psychological needs is critical for fostering intrinsic motivation. Studies by Kyewski and Krämer (2018) and García-Iruela et al. (2022) further confirm that more complex and interactive gamification designs yield stronger engagement effects.

Moreover, integrating metacognitive reading strategies into an engaging gamified context helped simplify complex cognitive processes and encouraged gradual mastery. Research has shown that student-centered environments, combined with autonomy and emotional engagement, significantly enhance strategy use and motivation (Dougherty, 1996; Jamshidi et al., 2013; Meniado, 2016; Arif et al., 2020). Feedback and reward systems further reinforced students' achievements and fostered a sense of progression.

In contrast to traditional methods often marked by low interactivity, the GIEB provided a dynamic, learner-centered experience that encouraged active participation and sustained motivation.

6.2 Mathematical reading comprehension

The GIEB intervention significantly improved students' mathematical reading comprehension, corroborating previous findings that gamification can positively influence academic performance by shaping learning behaviors and attitudes (Prados Sánchez et al., 2023; Li & Chu, 2021; Ndegwa et al., 2024). In contrast, Chen et al. (2020) cautioned that ineffective alignment between learning objectives and gamified components may reduce their impact on comprehension, underscoring the importance of meaningful integration.

The TL group showed no significant gains, consistent with studies suggesting that traditional metacognitive strategy instruction often lacks interactivity and immediate feedback, which may hinder student engagement and comprehension (Meniado, 2016; Pei, 2014; Arif et al., 2020). Boredom or emotional disengagement, commonly observed in passive instructional formats, may further impair outcomes (Shehzad et al., 2020).

In contrast, GIEB's design combined engaging game elements with clearly defined learning tasks. Features such as real-time feedback, virtual assistants, and reward mechanisms were strategically aligned with reading phases—supporting goal setting, monitoring, and reflection—thereby reinforcing comprehension (Chen et al., 2020). Moreover, the platform stimulated intrinsic motivation and positive emotions, which,

according to Fredrickson's (2001) broaden-and-build theory, can enhance cognitive engagement and learning.

Detailed feedback within GIEB not only clarified correct answers but also provided step-by-step reasoning (Tsai et al., 2020), helping students refine their understanding and identify gaps. These findings support the view that well-integrated gamified learning tools, when paired with metacognitive strategies, can effectively improve mathematical reading comprehension.

6.3 Performance in mathematical word problems

The GIEB significantly improved students' performance in mathematical word problems, likely due to its enhancement of mathematical reading comprehension. By embedding metacognitive strategies—prediction, monitoring, and reflection—into a gamified environment, students became better at identifying and analyzing problem-relevant information.

Gamified features such as real-time feedback, virtual assistants, and rewards supported motivation by fulfilling students' needs for autonomy, competence, and relatedness, in line with Self-Determination Theory (SDT). Autonomy was reinforced through learner control; competence was developed via positive feedback and guided support; and relatedness was encouraged through a positive, interactive environment that fostered emotional engagement and achievement.

In contrast, the TL group showed no significant improvement, echoing prior studies suggesting that traditional instruction may lack the motivational and cognitive support needed for success in solving word problems (Strohmaier et al., 2022; Vilenius-Tuohimaa et al., 2008). This finding is consistent with Sun-Lin and Chiou (2019), who emphasized the value of gamified design in creating engaging, cognitively stimulating learning environments.

Nonetheless, these results should be interpreted with caution. The novelty of the GIEB may have temporarily elevated student engagement (Clark, 1983; Rouse, 2013), and individual differences in technological proficiency could have influenced outcomes (Teo, 2009; Huang et al., 2021). Despite standardized procedures, such factors may have introduced subtle bias. Future studies should employ delayed post-tests and control for prior digital experience to better evaluate the intervention's long-term effects (Sailer & Homner, 2020a, b).

7 Limitation and recommendation

This study provides valuable insights into the potential of gamified interactive e-books (GIEB) incorporating metacognitive reading strategies to enhance students' performance in mathematical word problems. However, interpreting the findings requires consideration of its limitations, particularly regarding potential biases and constraints in generalizability.

The study involved a relatively small sample size of only 60 fifth-grade students from a single public elementary school in Hefei, Anhui Province, China. This limited sample size raises concerns about the statistical power of the findings, as small

samples may not capture the variability in the population. Additionally, the specific research context that limited to one school in a particular region might introduces potential selection bias. The convenience sampling method used may inadvertently reflect characteristics unique to the chosen school, such as teaching practices, student demographics, or local curriculum influences. These factors limit the generalizability of the findings to broader populations of Chinese elementary school students or students in other educational systems. Future studies should aim to address this limitation by including larger, more diverse samples, encompassing different regions, school types, and socioeconomic backgrounds, to enhance the external validity of the results.

Additionally, the potential influence of novelty effects and individual differences in students' technological proficiency may have introduced bias into the findings, despite efforts to standardize the intervention (Clark, 1983; Rouse, 2013; Teo, 2009; Huang et al., 2021). Future studies should consider controlling for these factors to enhance internal validity (Clark, 1983; Sailer & Homner, 2020a, b).

The current GIEB design assumes that gamified elements will elicit uniform responses across all students. However, significant variations in students' learning preferences, prior experiences with technology, and intrinsic motivational factors could lead to differential effects. This assumption may introduce bias, as the effectiveness of gamification could be overestimated or underestimated depending on how well it aligns with individual student needs. Future research could address this limitation by incorporating adaptive gamification techniques, which tailor game mechanisms to individual characteristics and learning profiles, ensuring more equitable and effective learning outcomes.

While this study focused on three metacognitive reading strategies—prediction, self-questioning, and reflection—at different stages of the GIEB, it does not account for the full range of metacognitive strategies available. The selection of these specific strategies may have introduced a form of selection bias, as other potentially effective strategies were excluded. This constraint might limit the instructional scope of the GIEB. Future research could explore integrating a broader array of metacognitive strategies to create more comprehensive and flexible instructional designs.

The study primarily examined the short-term effects of the GIEB intervention over a two-month period. While the findings demonstrate immediate benefits in terms of reading motivation, mathematical reading comprehension, and mathematical word problem-solving performance, the long-term sustainability of these improvements remains uncertain. The absence of delayed post-tests or extended intervention periods constrains the study's ability to determine the durability of these outcomes. Future research should address this limitation by extending the study duration or incorporating follow-up assessments to evaluate the retention of gains over time and their potential transfer to other academic contexts.

These limitations highlight the need for caution when interpreting the findings of this study. While the results are promising, addressing these biases and constraints in future research will provide a more deep understanding of the effectiveness of gamified interactive e-books and metacognitive reading strategies, thereby enhancing their applicability and impact on diverse educational settings.

8 Conclusion

This study explored the impact of gamified interactive e-books (GIEB) incorporating metacognitive reading strategies on elementary school students' reading motivation, mathematical reading comprehension, and performance in mathematical word problems. The results show that the combination of staged guidance strategies and gamification elements in GIEB not only significantly improved students' mathematical reading comprehension but also enhanced their performance in mathematical word problems by stimulating intrinsic motivation for learning.

Specifically, metacognitive reading strategies played a key role in extracting and analyzing mathematical problem information, while the gamified design, through mechanisms such as immediate feedback, virtual assistants, and rewards, fulfilled students' needs for autonomy, competence, and relatedness. This increased the enjoyment and engagement of learning, thereby optimizing students' learning experiences and outcomes. This finding further validates the applicability of Self-Determination Theory (SDT) in educational contexts, demonstrating that meeting basic psychological needs can significantly boost students' intrinsic motivation and academic performance.

The findings of this study align with the United Nations Sustainable Development Goal 4 (Quality Education), providing an effective practical approach to improving educational quality and learning outcomes. It also offers an innovative solution to achieving educational equity, particularly in promoting students' understanding and application of mathematical knowledge.

In conclusion, this study highlights that while metacognitive strategies are widely recognized in language learning, their potential in mathematics education, particularly through gamified approaches, represents an innovative contribution to the field. This study provides an innovative practice in the field of mathematics education, indicating that digital tools combining metacognitive strategies with gamification elements can effectively enhance students' learning motivation and academic performance. Not only does it offer theoretical support based on SDT for educational practice, but it also contributes significant value toward achieving educational equity, improving educational quality, and advancing sustainable development goals.

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Data availability The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Declarations

Ethics approval and informed consent All studies involving human participants were reviewed and approved by the Ethics Committee of the School of Design, Anhui University of Engineering, and strictly adhered to the ethical principles of the Declaration of Helsinki. All participants provided informed consent prior to formal participation in this study.

Consent for publication Consent for publication was obtained from the participants.

Competing interests There is no potential conflict of interest in this study.

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